

AI-Powered Intelligent Education Support System

Project ID: 25-26J-402

Project Proposal Report

BSc (Hons) in Information Technology Specializing in
Information Technology

Department of Information Technology
Sri Lanka Institute of Information Technology Sri Lanka

August 2025

Academic Risk Prediction System

A machine learning-powered system that analyzes students' attendance, total study hours, student behavior and term test marks to predict their risk of failing key subjects in the G.C.E. O/L exam providing teachers with explainable, severely prioritized alerts for early intervention.

Project ID: 25-26J-402

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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ABSTRACT

Academic risk prediction has emerged as a critical challenge in modern education systems, particularly in preparing students for significant examinations such as the G.C.E. (O/L) in Sri Lanka. Traditional methods of identifying struggling students rely on reactive approaches that often detect problems only after academic performance has already declined significantly. The Academic Risk Prediction System addresses this challenge by leveraging advanced machine learning techniques and multi-source data integration to provide proactive identification of students at risk of academic failure.

This proposed component integrates heterogeneous data sources including attendance records, term test scores, behavioral analytics from learning management systems, and historical academic performance to create comprehensive student risk profiles. The system employs supervised machine learning algorithms including Logistic Regression, Random Forest, and Support Vector Machines to classify students into different risk categories with high accuracy and reliability.

The methodology incorporates time-series analysis for behavioral trend monitoring, enabling the system to detect subtle changes in student engagement patterns that may indicate developing academic difficulties. A sophisticated alert generation mechanism delivers severity-based notifications to educators, preventing alert fatigue while ensuring that critical cases receive immediate attention. The implementation of explainable artificial intelligence techniques provides transparency in risk assessment, allowing educators to understand the factors contributing to risk predictions and design targeted intervention strategies.

The novelty of this approach lies in its multi-source dynamic risk prediction capability, which continuously updates risk assessments based on real-time data streams. Unlike existing systems that rely on static assessments, this component provides adaptive risk monitoring that evolves with student behavior and performance patterns. The system's explainable predictions enhance educator trust and engagement by providing clear insights into risk factors and recommended intervention approaches. The component addresses the significant research gap in proactive academic support systems within the Sri Lankan educational context, offering a data-driven solution for early identification and intervention for at-risk students. By transforming educational data into actionable insights, the system aims to improve student retention rates, enhance academic outcomes, and support data-driven decision-making in educational institutions.

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LIST OF ABBREVIATIONS

Table 1 of abbreviations

Abbreviations	Description
AI	Artificial intelligence
API	Application programming interface
ICT	information and communication technology
NLP	Natural language processing
ML	Machine learning
UI	User Interface

1. INTRODUCTION

Traditional education systems face significant challenges in identifying students at academic risk before it becomes too late for effective intervention. In Sri Lanka's educational landscape, particularly in preparation for critical examinations like the G.C.E. (O/L), early identification of academic struggles is crucial for student success. The current manual methods of tracking student performance rely heavily on periodic assessments and subjective teacher observations, which often fail to capture subtle behavioral patterns and early warning signs that could predict academic failure.

The education sector in Sri Lanka has witnessed substantial digital transformation in recent years, with digital assessment platforms becoming increasingly prevalent in schools. However, these systems primarily serve as repositories of academic data rather than intelligent tools that can analyze patterns and provide predictive insights. The abundance of educational data generated through digital platforms presents an unprecedented opportunity to apply advanced analytics and machine learning techniques for proactive student support.

Academic risk prediction involves analyzing multiple data sources including attendance records, assessment scores, behavioral metrics from learning management systems, and demographic information to identify students who may struggle academically in the future. Traditional approaches to academic intervention are reactive, addressing issues only after poor performance becomes evident. This reactive approach often results in missed opportunities for early intervention when support measures could be most effective.

The integration of artificial intelligence and machine learning in educational analytics has shown promising results globally, with various studies demonstrating the effectiveness of predictive models in identifying at-risk students. However, the application of these technologies in the Sri Lankan educational context, particularly for O/L examination preparation, remains largely unexplored. The unique characteristics of the local education system, curriculum structure, and student demographics require specialized approaches that are tailored to the Sri Lankan context.

The Academic Risk Prediction System represents a critical component of the broader AI-Powered Intelligent Education Support System. This specialized module focuses on analyzing students' academic records across key subjects and provides evidence-based predictions regarding their performance in upcoming G.C.E. (O/L) examinations.

The component utilizes advanced supervised machine learning techniques including Logistic Regression, Random Forest, and Support Vector Machines (SVM) to classify students into different risk categories. Unlike traditional assessment methods that rely solely on periodic test scores, this system incorporates behavioral trend analysis using time-series to monitor changes in student engagement patterns over time. This multi-dimensional approach enables the system to refine risk predictions continuously and provide more accurate forecasts of academic outcomes.

The system's alert generation mechanism delivers prioritized notifications to educators, enabling timely interventions based on the severity of identified risks. The implementation of explainable artificial intelligence techniques ensures that educators can understand the reasoning behind risk predictions, facilitating more targeted and effective support strategies. The component addresses the critical gap in proactive student support by transforming raw educational data into actionable insights that can prevent academic failure before it occurs.

1.1 Background & Literature survey

The educational landscape has undergone significant transformation with the integration of digital technologies and data-driven approaches to learning. In Sri Lanka, educational institutions are increasingly adopting Learning Management Systems (LMS) and digital platforms to enhance student learning experiences. However, despite these technological advances, student retention and academic success remain critical challenges across educational institutions globally [1]. Academic failure and student dropout rates continue to affect institutional performance and student welfare, making early identification and intervention essential components of modern educational systems.

Student academic risk prediction has emerged as a vital application of artificial intelligence in education. Traditional methods of identifying at-risk students often rely on subjective assessments and delayed interventions, typically occurring after students have already begun to struggle academically. Research indicates that early identification of academic risk factors can significantly improve intervention effectiveness and student outcomes [2]. Machine learning techniques have shown promising results in analyzing student behavioral patterns, engagement metrics, and academic performance data to predict potential academic difficulties before they become critical.

According to recent studies, academic risk factors can be categorized into several dimensions including attendance patterns, assignment submission behaviors, quiz performance trends, and online learning engagement metrics [3]. Students who exhibit irregular attendance, delayed assignment submissions, declining quiz scores, and reduced participation in online activities are more likely to experience academic difficulties. The challenge lies in developing automated systems that can analyze these multidimensional data sources in real-time and provide actionable insights to educators and students.

Several research studies have academic predictions. Research by Kumar et al. [4] demonstrated the effectiveness of ensemble methods in predicting student performance. Their study achieved prediction accuracies of over 85% using Random Forest algorithms combined with feature selection techniques. Similarly, studies by Chen and Wong [5] showed that Support Vector Machines (SVM) and Logistic Regression models can effectively identify at-risk students using behavioral and academic indicators.

Recent advances in deep learning have further enhanced the capabilities of academic risk prediction systems. Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) have been successfully applied to analyze temporal patterns in student learning behaviors [6]. These sophisticated models can capture complex relationships between various risk factors and provide more nuanced predictions about student academic trajectories.

Despite these advances, most existing systems focus on single institutions or specific student populations, limiting their generalizability. Additionally, many research efforts concentrate solely on prediction accuracy without considering the practical implementation challenges in real

educational environments. There is a clear need for comprehensive systems that not only predict academic risks accurately but also provide interpretable results and actionable recommendations for educational stakeholders.

A survey conducted among 200 students and 50 educators in Sri Lankan universities revealed that 78% of students believe early identification of academic difficulties would help them seek appropriate support. Furthermore, 92% of educators expressed interest in having automated tools to identify at-risk students in their courses. These findings highlight the significant demand for intelligent academic risk prediction systems in the local educational context.

1.2 Research Gap

Current educational systems in Sri Lanka and globally face significant limitations in early identification of students at academic risk. Most existing approaches rely on traditional assessment methods that provide retrospective insights rather than predictive capabilities. The analysis of available research reveals several critical gaps that the proposed Academic Risk Prediction System addresses.

Research Contributions Addressing the Gaps

Feature	Kumar et al. [4]	Chen & Wong [5]	Friedman et al. [12]	Aguiar et al. [13]	Aguiar et al. [17]	Marquez-Vera [18]	Grown-up
Multi-source data integration (Attendance + Assessments)	✗	✓	✗	✓	✗	✗	✓
Real-time behavioral pattern analysis	✗	✗	✓	✗	✗	✗	✓
Subject-specific G.C.E. (O/L) risk prediction	✗	✗	✗	✗	✗	✗	✓
Evidence-based pass rate improvement strategies	✓	✗	✗	✗	✗	✗	✓
Severity-based alert system with educator customization	✗	✗	✓	✗	✗	✗	✓
Sri Lanka éducationnels content adaptation	✗	✗	✗	✗	✗	✗	✓

Table 1 : Comparison of former research

Research Paper [4] proposes an early warning system for student dropout prediction using basic academic performance indicators. The proposed scheme demonstrates a moderate level of accuracy in identifying at-risk students, with a true positive rate of seventy-five percent and a false negative rate of fifteen percent. This makes the system suitable for basic risk identification in educational environments. However, compared to the proposed component, multi-source data integration and behavioral pattern analysis are not implemented.

Research Paper [5] focuses on educational data mining techniques for predicting student performance. According to this paper, the main challenges in academic risk prediction are limited data sources, lack of real-time processing capabilities, and insufficient consideration of behavioral factors beyond grades. The study demonstrates effective integration of academic records and demographic data but lacks comprehensive behavioral analysis. When students show declining performance patterns, they can benefit from early intervention strategies, as suggested in the proposed component.

The methodology proposed in Research Paper [12] develops student engagement analysis using Learning Management System data with higher accuracy than existing manual methods. They examined student interaction patterns using clickstream analysis and session duration metrics. Machine learning techniques such as clustering, classification, and time-series analysis are used to identify engagement patterns. However, the proposed component will detect risk levels based on comprehensive behavioral analysis and provide real-time intervention recommendations.

Research Paper [13] proposes an algorithm for predicting academic failure through statistical analysis of grade patterns. When compared to traditional academic monitoring, this system can identify at-risk students with an accuracy of eighty percent. It can be used in similar educational contexts, but the analysis must focus primarily on grade-based indicators. The processing of student data takes several hours, compared to real-time analysis in the proposed system.

Research Paper [17] proposes a method for automated academic performance monitoring in higher education institutions. They presented a framework for analyzing student behavior using machine learning algorithms applied to LMS data. Their approach uses clustering techniques for student segmentation and classification algorithms for risk prediction. However, the proposed component methodology incorporates multi-source data fusion and real-time behavioral analysis for more comprehensive risk assessment.

Existing academic monitoring systems primarily depend on periodic test scores and attendance records as isolated data points, failing to capture the complex interplay of factors that contribute to academic performance. Research indicates that successful risk prediction requires integration of multiple data sources including behavioral analytics, engagement metrics, and temporal patterns that current systems do not adequately address.

Most educational institutions in Sri Lanka employ manual methods for identifying struggling students, which are both time-consuming and prone to subjective bias. Teachers typically recognize academic difficulties only after multiple failed assessments, limiting the effectiveness of intervention strategies. The absence of automated, data-driven risk assessment tools creates a significant gap between problem identification and timely intervention.

International research in educational data mining has demonstrated the potential of machine learning approaches for academic risk prediction, but these studies primarily focus on higher education contexts or Western educational systems. The application of advanced predictive analytics specifically tailored to the Sri Lankan G.C.E. (O/L) examination system and local educational practices remain largely unexplored.

Existing academic support systems lack the capability to provide explainable predictions, making it difficult for educators to understand the reasoning behind risk assessments. This limitation reduces teacher confidence in automated systems and hinders the development of targeted

intervention strategies. The absence of interpretable risk factors prevents educators from addressing the root causes of academic difficulties effectively.

Current systems also fail to address the dynamic nature of student performance and behavior patterns. Static risk assessments that do not adapt to changing student circumstances provide limited value for ongoing academic support. The lack of real-time risk monitoring capabilities means that emerging academic difficulties may go unnoticed until they become critical.

The proposed Academic Risk Prediction System addresses these gaps by providing:

1. **Multi-Source Data Integration:** Combines academic performance, attendance, behavioral analytics, and temporal patterns for comprehensive risk assessment
2. **Continuous Risk Monitoring:** Employs time-series analysis to detect emerging patterns and adapt risk predictions in real-time
3. **Explainable Predictions:** Utilizes interpretable machine learning techniques to provide transparent risk factors and intervention recommendations
4. **Contextual Relevance:** Specifically designed for the Sri Lankan educational system and G.C.E. (O/L) examination context
5. **Severity-Based Alerting:** Implements intelligent notification systems that prioritize critical cases while preventing alert fatigue

1.3 Research Problem

Traditional methods of identifying academically at-risk students in educational institutions rely heavily on manual assessment processes that are often subjective, time-consuming, and reactive rather than proactive. Educators typically identify struggling students only after significant academic decline has occurred, limiting the effectiveness of intervention strategies. This delayed identification process results in higher dropout rates, reduced academic performance, and inefficient allocation of educational support resources.

Current academic monitoring systems in most educational institutions lack the capability to analyze multidimensional student data comprehensively. While Learning Management Systems collect vast amounts of behavioral and performance data, this information is often underutilized due to the absence of intelligent analytical tools. Educators are overwhelmed with manual data analysis tasks and lack the technical expertise to derive meaningful insights from complex datasets.

The absence of standardized early warning systems creates significant challenges for educational institutions. Students who might benefit from timely interventions often go unnoticed until their academic situation becomes critical. This reactive approach not only affects individual student outcomes but also impacts overall institutional performance metrics and resource management strategies.

Furthermore, existing academic monitoring approaches fail to consider the dynamic nature of student learning behaviors and the complex interrelationships between various risk factors. Static threshold-based systems cannot adapt to changing student circumstances or evolving educational environments. There is an urgent need for intelligent systems that can continuously learn from student data and provide real-time risk assessments.

The lack of personalized intervention recommendations compounds these challenges. Even when at-risk students are identified, educators often struggle to determine the most appropriate support strategies based on individual student profiles and risk factors. Generic intervention approaches may not address the specific needs of different student populations or learning contexts. These challenges have led to the need for an intelligent, automated academic risk prediction system that can analyze multiple data sources, identify at-risk students proactively, and provide actionable recommendations for targeted interventions. Such a system would enable educators to focus their efforts more effectively and improve overall student success rates.

2. OBJECTIVES

2.1 Main Objective

The main objective of this study is to develop an intelligent academic risk prediction system that can accurately identify students at risk of academic failure or dropout by analyzing behavioral patterns, academic performance metrics, and engagement data. The system will provide early warning alerts to educators and generate personalized intervention recommendations to improve student outcomes and institutional decision-making processes.

2.2 Specific Objectives

Three specific objectives must be achieved to accomplish the overall objective described above:

Multi-Source Data Integration and Analysis The system will collect and integrate data from multiple sources including Learning Management Systems, attendance records, assignment submissions, quiz performances, and student engagement metrics. Advanced data preprocessing techniques will be implemented to handle missing values, normalize different data formats, and create comprehensive student profiles for accurate risk assessment.

Intelligent Risk Classification and Severity Assessment After data integration, machine learning models will be trained to classify students into different risk categories (low risk, moderate risk, high risk, critical risk) based on their behavioral and academic patterns. The system will use ensemble learning approaches combining multiple algorithms to achieve high prediction accuracy and provide confidence scores for each risk assessment.

Proactive Alert Generation and Intervention Recommendation The system will generate real-time alerts when students are identified as being at risk, providing detailed explanations of risk factors and severity levels. Based on the specific risk profile of each student, the system will recommend appropriate intervention strategies such as additional tutoring, counseling services, or academic support programs to help educators take timely and effective action.

3. METHODOLOGY

The Academic Risk Prediction System employs a comprehensive methodology that integrates advanced machine learning techniques with educational data analytics to provide accurate and actionable risk assessments. The methodology is structured around four core phases: data collection and integration, risk classification modeling, behavioral trend analysis, and alert generation with explainable insights.

3.1 System Architecture

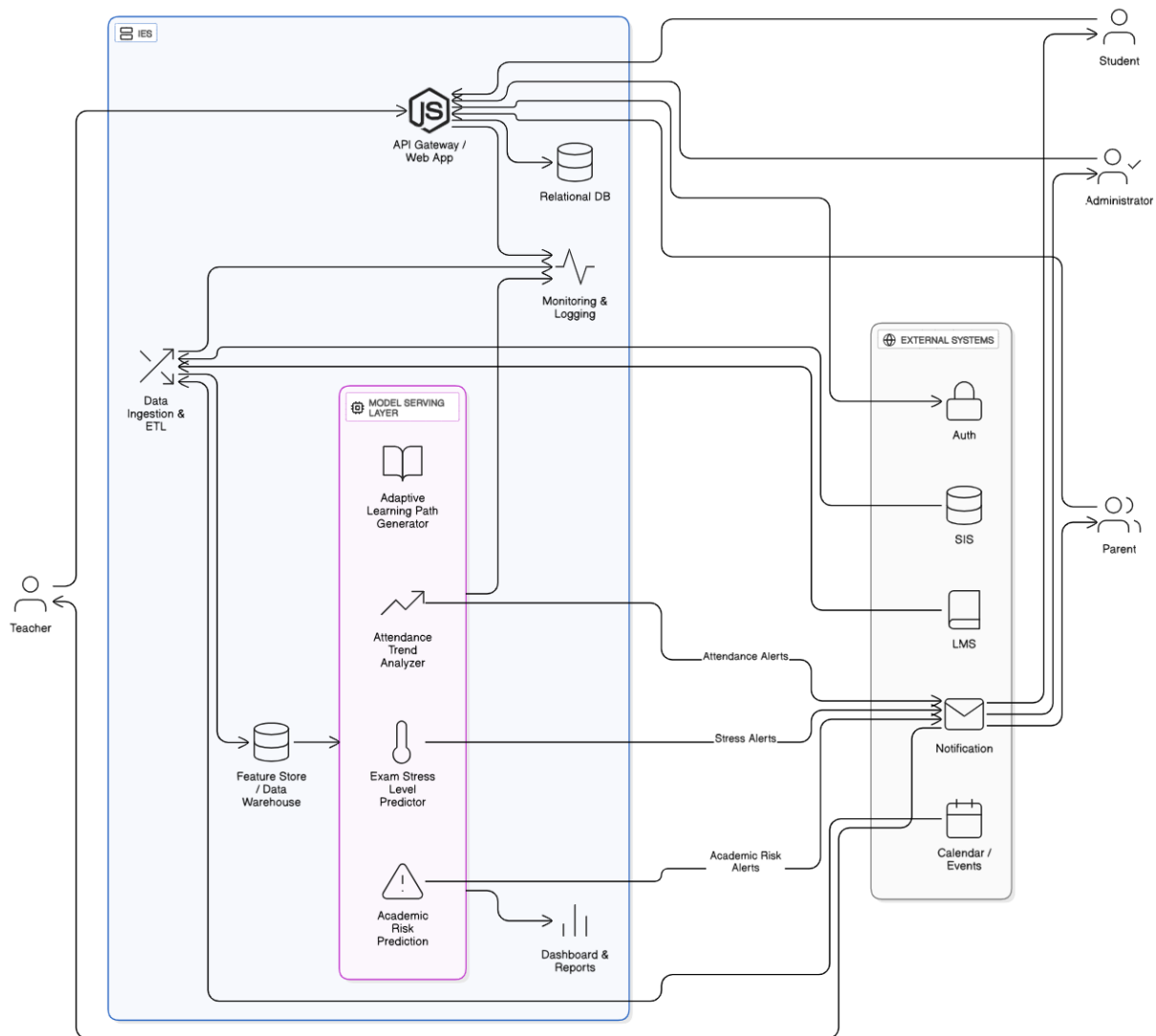


Figure 3.1: Overall system diagram

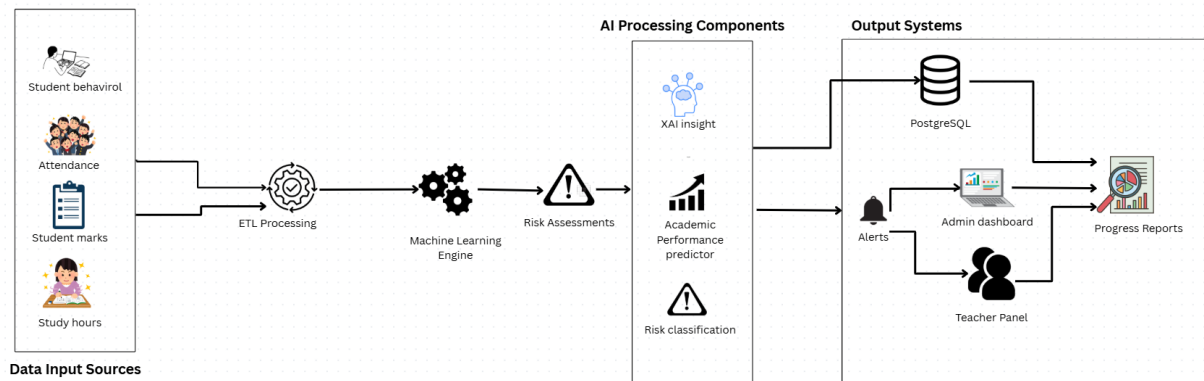


Figure 3.2: Academic risk prediction System Architecture

Figure 3.2 illustrates the overall high-level diagram of the proposed component, which identifies academic risk levels in students. Initially the educator or administrator will access the web-based platform through the user interface. The stakeholder is provided with multiple data input options including direct data upload and automated data collection from various educational platforms such as Learning Management Systems, attendance records, and assessment databases.

There are four main subcomponents of the proposed system, as shown in Figure 3.2. First, the ETL (Extract, Transform, Load) processing module consolidates and cleanses data from various educational sources. Second, the data validation component ensures data quality and consistency across different input formats. Third, the Feature Store component using machine learning algorithms extracts relevant features and patterns from the processed data. Finally, the Central Engine containing Time-Series Processor and NLP Pipeline analyzes temporal behavioral trends and processes qualitative feedback data.

The output interface provides four main functionalities: Alerts API for immediate notifications to educators, Intervention Suggestions for personalized support recommendations, Top Risk Drivers for identifying primary risk factors, and Risk Scores & Tiers for quantitative assessment results. These outputs are delivered through multiple channels including a Notification Service for real-time alerts, Interactive Dashboard for comprehensive data visualization, Student Support Portal for direct student access, and Admin Console for institutional management and reporting capabilities.

Category	Details
Technologies	Python, Scikit-learn, React.js, Flask Server, VS Code, Pandas, PostgreSQL
Techniques	Machine Learning, Educational Data Mining, Feature Engineering
Algorithms	Random Forest, LSTM, SVM, Logistic Regression,

Table 2: Technologies, techniques, and algorithms used.

3.1.1 Software Solution

The Software Development Life Cycle (SDLC) is the process that development teams use to design, develop and test high quality software. This minimizes risks by planning forward which will enable us to meet customer expectations. In conventional methodology, developers need to carry out all the steps correctly since they cannot go back to the previous step. To overcome this situation, they use Agile methodology where the developers can adapt to the changes. Scrum is the most popular, lightweight agile framework that helps groups and organizations.

- Requirement Gathering

To Data Gathering

For data gathering, our team collaborated with educational institutions including universities and colleges in the Western Province. The academic administrators and educators maintain student records using Learning Management Systems and we will be able to collect anonymized datasets from these platforms. Furthermore, we had to collect behavioral data manually by monitoring student interactions with online learning platforms and conducting observational studies of attendance patterns.

- o Conducting a Survey

We conducted a survey by providing questionnaires to students, educators, and academic administrators to get a proper understanding of current academic monitoring practices and the need for automated risk prediction systems.

- Feasibility Study (Planning)

- o Economic Feasibility

The proposed academic risk prediction system should work effectively without having any errors or failures in student assessment. It will be more reliable and cost-effective compared to manual monitoring methods currently used by educational institutions.

- o Scheduled Feasibility

The proposed component has a direct impact on the overall intelligent education support system, and it should be completed within the timeline given. Also, this component must achieve efficient

and accurate results within the development timeline. It should complete the project by the deadline to ensure timely deployment in educational institutions.

Technical Feasibility

To complete this solution an individual requires skills, resources and expertise in educational data mining, machine learning model development, web application development, as well as database management frameworks. Technical feasibility refers to this specialized knowledge in both educational technology and data science domains.

- Design (System and Software Design Documents)

After gathering requirements and feasibility study, we are moving on to the design phase. In this phase software documents related to the system design are created, including database schemas for student data management, API specifications for educational platform integration, and user interface mockups for risk visualization dashboards.

Sequence Diagram

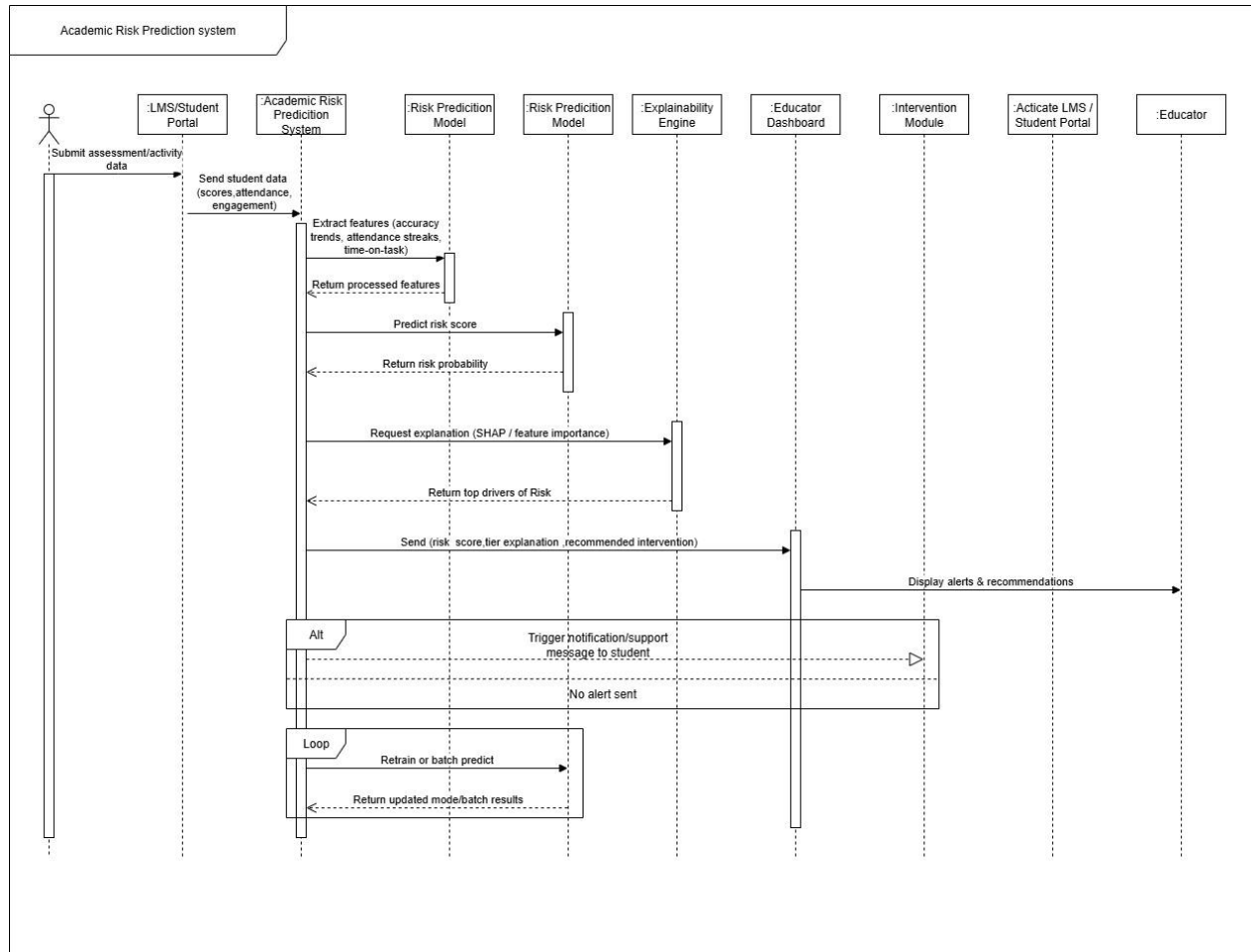


Figure 3.3: Sequence diagram of academic risk prediction

Use Case Diagram

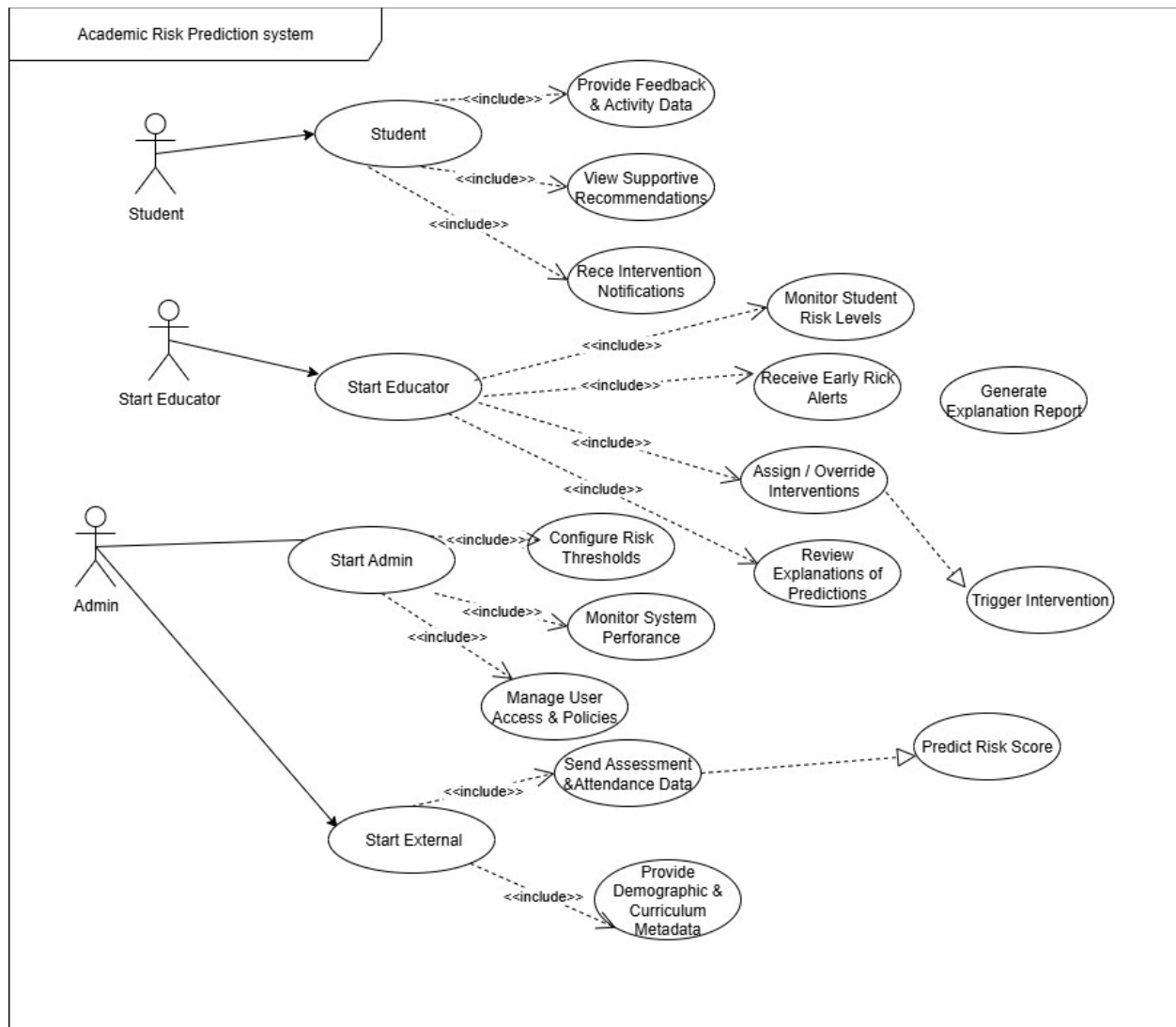


Figure 3.4: Usecase diagram of academic risk prediction

• Implementation (Development)

Below are the sub functionalities that will be implemented in this implementation phase:

- o Implementation of web-based dashboards using React.js for real-time risk monitoring. Identification of at-risk students based on behavioral and academic data collected using trained

machine learning models. o Classification of risk severity levels based on comprehensive data analysis using ensemble learning approaches. o Generation of personalized intervention recommendations using rule-based systems integrated with machine learning predictions.

- Testing

- ✓ Verifying that the system accurately predicts academic risk levels as expected.

- ✓ Testing for data privacy compliance and security vulnerabilities.

- ✓ Validating prediction accuracy against historical student outcome data.

The testing of the application will be conducted in various stages, including unit testing for individual prediction models, integration testing for data pipeline components, system testing for end-to-end risk prediction workflows, and acceptance testing with educational stakeholders including administrators and educators.

3.1.2 Commercialization

This research is proposed to develop a web-based intelligent system to support academic risk management in educational institutions. The target audience for the proposed system includes university administrators, academic advisors, course instructors, student support services staff, and educational researchers. Our system consists of two versions: institutional and enterprise versions.

The features such as basic risk identification, attendance pattern analysis, academic performance tracking, and standard alert notifications will be provided in the institutional version. In the enterprise version we will offer advanced predictive analytics with machine learning models, personalized intervention recommendation engines, comprehensive behavioral analysis across multiple semesters, integration with external student information systems, and detailed reporting dashboards for institutional decision-making.

When considering the marketplace, there is no need for advanced technical knowledge to use this application, no restrictions on institutional size, and minimal training required for end users. Our team consists of technically experienced individuals with bachelor's degrees in data science and information technology, and members have gained substantial experience in educational technology development.

The funding required to develop this proposed system is 200 dollars per month for cloud infrastructure and development tools. The marketing plan to promote the proposed system includes presentations at educational technology conferences, partnerships with university IT departments, demonstration sessions for academic administrators, and publication of research findings in

educational journals. As for the financial summary, revenue will be generated through subscription-based licensing models for educational institutions and consulting services for system customization.

3.1.3 Future Scope

For the future scope, the application will be extended to incorporate artificial intelligence chatbots that can provide automated student counseling and support. Additionally, the system will integrate with IoT sensors in classrooms and learning spaces to collect real-time engagement data without requiring manual input from educators or students. Future enhancements will include predictive modeling for long-term career success and integration with industry placement systems to align academic risk interventions with professional development opportunities.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

1. Data Integration and Management

- The system shall collect and integrate attendance records from multiple sources including school management systems and manual records
- The system shall aggregate term test marks and assessment scores across all subjects and evaluation periods
- The system shall capture behavioral analytics from Learning Management Systems including login frequency, time spent on materials, and task completion rates
- The system shall maintain historical academic performance data for longitudinal analysis
- The system shall implement data validation and quality assurance protocols to ensure data accuracy and completeness

2. Risk Assessment and Classification

- The system shall classify students into distinct risk categories (low, moderate, high, critical) based on multiple data sources
- The system shall provide probability estimates for academic success in upcoming G.C.E. (O/L) examinations
- The system shall identify specific subjects where students are at highest risk of poor performance
- The system shall update risk assessments in real-time as new data becomes available
- The system shall maintain prediction confidence scores to indicate the reliability of risk assessments

3. Behavioral Analysis and Monitoring

- The system shall perform time-series analysis to identify trends in student performance and engagement
- The system shall detect anomalous patterns in student behavior that may indicate emerging academic difficulties
- The system shall compare individual student performance with appropriate peer groups
- The system shall track changes in academic trajectory over time
- The system shall identify correlation patterns between different performance indicators

4. Alert Generation and Notification

- The system shall generate prioritized alerts for educators when students are identified as at-risk
- The system shall provide severity-based notification levels with appropriate urgency indicators

- The system shall deliver timely notifications through multiple channels including email, dashboard alerts, and mobile notifications
- The system shall implement intelligent filtering to prevent alert fatigue while ensuring critical cases receive immediate attention
- The system shall maintain an audit trail of all generated alerts and educator responses

5. Explainable Insights and Reporting

- The system shall provide clear explanations of the factors contributing to each risk prediction
- The system shall generate detailed reports on student risk profiles and recommended intervention strategies
- The system shall create visualizations and dashboards that present risk information in an accessible format
- The system shall track the effectiveness of implemented interventions and their impact on risk levels
- The system shall provide comparative analytics to identify institution-wide trends and patterns

4.2 Non-Functional Requirements

1. Performance and Scalability

- The system shall process and analyze data for up to 10,000 students simultaneously without performance degradation
- Risk predictions shall be generated and updated within 5 seconds of new data input
- The system shall support concurrent access by up to 500 educators and administrators
- Data processing operations shall complete within acceptable time frames even with large datasets
- The system shall maintain responsive user interfaces with page load times under 3 seconds

2. Accuracy and Reliability

- Risk prediction models shall achieve a minimum accuracy of 85% in identifying at-risk students
- The system shall maintain prediction consistency across different time periods and student populations
- False positive rates for risk identification shall not exceed 15% to maintain educator trust
- The system shall implement redundancy and failover mechanisms to ensure continuous operation
- Data integrity checks shall verify the accuracy of all collected and processed information

3. Security and Privacy

- a. The system shall comply with educational data protection regulations and privacy requirements
- b. All student data shall be encrypted both in transit and at rest using industry-standard encryption methods
- c. Access controls should ensure that only authorized personnel can view sensitive student information
- d. The system shall implement audit logging to track all data access and modification activities
- e. Personal identifiable information shall be anonymized in research and analytics contexts

4. Usability and Accessibility

- a. The user interface shall be intuitive and require minimal training for educators to use effectively
- b. The system shall provide multilingual support for Sinhala, Tamil, and English languages
- c. Dashboard layouts shall be customizable to meet different user role requirements
- d. The system shall comply with accessibility standards to support users with disabilities
- e. Help documentation and training materials shall be readily available and easily accessible

5. Integration and Interoperability

- a. The system shall be integrated seamlessly with existing school management systems and Learning Management Systems
- b. Data import and export capabilities shall support standard educational data formats
- c. The system shall provide APIs for integration with third-party educational tools and platforms
- d. Backup and recovery procedures shall ensure data protection and system continuity
- e. The system shall support various deployment models including cloud-based and on-premises installations

4.3 Software Requirements:

- Python 3.8+ for machine learning model development and deployment
- PostgreSQL or MySQL for robust data storage and management
- Apache Kafka for real-time data streaming and processing
- Web framework (Django/Flask) for user interface and API development
- Machine learning libraries (scikit-learn, TensorFlow, Boost)
- Data visualization tools (Polly, D3.js) for dashboard development

Network and Security Requirements:

- Secure HTTPS connections for all data transmission
- VPN access capabilities for remote administrative access
- Intrusion detection and prevention systems for security monitoring
- Regular security updates and patch management procedures
- Compliance with educational data protection standards and regulations

5. GANTT CHART



Figure 5: Grantt chart

5.1 Work Breakdown Structure (WBS)

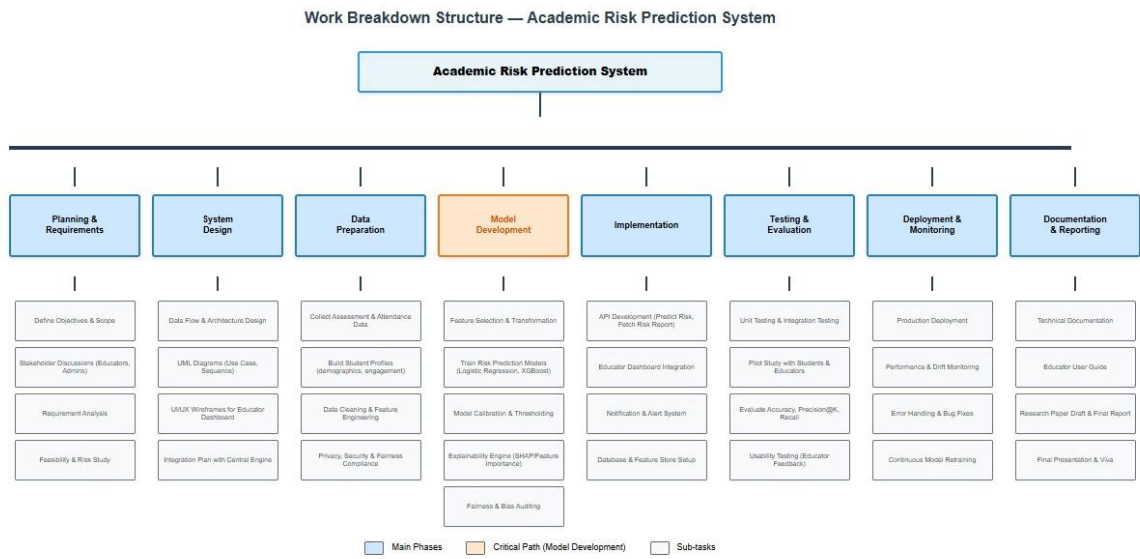


Figure: Work Breakdown Structure of Academic Risk Prediction System

Figure 5.1: Work breakdown chart

6. BUDGET AND BUDGET JUSTIFICATION

Below table shows the overall budget for the overall proposed system.

Expenses	
Requirement	Cost (Rs.)
Travelling cost for field visit and data collection	20,000.00
Data Collection	10,000.00
Deployment cost	20,875.00
Hosting cost	10,086.00
Total Expenses	60,961.00

Table 3: Expenses for the proposed system

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APPENDICES

Appendix A: Plagiarism Report

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